

Meli — 7-Month Budget Plan (Jun 1 – Dec 31, 2026)

Rev 7. iFlytek pricing corrected against official pricing docs at global.xfyun.cn/doc/platform/pricing.html. Actual pay-as-you-go rate is **USD 0.003/call** ($\approx$ HKD 0.023, $\approx$ ¥0.021 RMB) — half of the conservative HKD 0.044 estimate used in rev 6. Resource packs available but **not cost-effective at our scale** (Pack A at USD 150/year breaks even at $\sim$ 50K calls/year; 500-user scenario uses only $\sim$ 7K calls in 4 months). PAYG remains the right choice.

Rev 6. iFlytek pronunciation grading is now budgeted at **standard pay-as-you-go rates**. No free-tier assumption — if HKUST does qualify for the enterprise free quota, that becomes upside, not baseline.

Rev 5. (a) Resend removed — transactional email now goes through HKUST's institutional SMTP relay (free). (b) User-count scenarios capped at 500 (highest-likely target is 200). (c) Instructor-to-student ratio updated to **1:40** (was 1:25), which lowers instructor headcount and trims doc-processing share of variable AI cost.

Rev 4. Chinese-only deployment — pronunciation grading uses **iFlytek (讯飞)** Speech Evaluator, served entirely within the free enterprise tier (200K calls / 3 months). Includes streaming grading of single characters, words, sentences, and **long passages** under one call meter.

Rev 3. Pronunciation grading is a **secondary feature**, not a primary one — per-user attempt counts reduced from $\sim$ 16.5/mo to $\sim$ 3.7/mo.

Rev 2. (a) Whisper / lecture-audio transcription removed from scope; (b) Railway pricing corrected — Pro plan fee is a credit floor, not additive; (c) primary LLM switched to **DeepSeek V4 Flash** ($\sim$ 80% cheaper output than V3.2).

Context

Meli (HKUST $\times$ PSC anniversary showcase) goes live in **September 2026** and runs through end of December. Three months prior (**June – August**) are the development / closed-staging window. This budget covers the full 7 months and reports all figures in **HKD** with USD/RMB references where the upstream vendor prices in those currencies.

The codebase already wires every paid surface that will be active during the showcase (LLM, embeddings, VLM, iFlytek pronunciation, R2, Vercel, Railway). Transactional email (verification + reset) goes through HKUST's institutional SMTP relay, which is free for institutional projects. The five "showcase narrative" features the user mentioned — **placement test (5-group division), mid-semester survey, instructor analytics dashboard, department reports, weak-point personalized quiz** — are **additive on top of the existing AI surface**, not replacements. They are not yet coded (no specs in `docs/superpowers/`), so they are modeled as forward-looking line items.

User answers driving the model: - **User mix:** mostly students, $\sim$ 1 instructor per 40 students. - **Pronunciation grading:** Chinese-only, served by **iFlytek (讯飞)** at pay-as-you-go rates (no free-tier dependency). Development through Aug, live from Sep. - **Lecture-audio transcription:** out of scope. - **Email delivery:** HKUST institutional SMTP relay (free). - **User-count cap:** 500 maximum; 200 is the realistic upper bound for the showcase. - **Safety margin:** +25% applied to the headline subtotal.

FX assumptions: **1 USD $\approx$ 7.78 HKD, 1 USD $\approx$ 7.10 RMB** (Q2 2026 reference rates).

1. Cost-Surface Inventory

All citations are repo-relative paths.

#	Surface	Files	Trigger	Pricing model
1	LLM generation — quiz / flashcard / summary / pronunciation set / revision	services/generator.py:296 , :445 , :394 , :686	Per user/instructor action	DeepSeek V4 Flash primary, Gemini 2.5 Flash Lite fallback (OpenRouter)
2	Concept extraction	services/concept_extraction.py:35	Once per uploaded document (samples up to 200 chunks)	Same as #1
3	Concept tagging	services/concept_tagger.py:72	Once per chunk on doc upload + inheritance ×0.7 on artifact gen	Same as #1
4	Syllabus parsing	api/syllabus.py:163	Once per syllabus upload	Same as #1
5	Embeddings (chunks + queries + concepts)	services/embedder.py:27	Per chunk on upload + 1 per RAG query	openai/text-embedding-3-large via OpenRouter
6	VLM figure captioning (PDF/PPTX figures)	services/vlm.py:176 , services/parser.py:136	Per figure on doc upload, capped via enable_figure_captions / page_rescue_max_pages=40	Gemini 2.5 Flash (OpenRouter)
7	VLM page-rescue OCR (scanned PDF pages)	services/vlm.py:203 , services/parser.py:211	Per low-text page (<30 words), hard-capped at 40 pages	Same as #6
8	iFlytek (讯飞) Chinese pronunciation grading (single char / word / sentence / long passage)	services/speech.py:45	Per student pronunciation submission	iFlytek Speech Evaluator (ISE) — pay-as-you-go at USD 0.003/call (≈ HKD 0.023, ≈ ¥0.021 RMB), per official pricing docs
9	Object storage	services/storage.py	Document upload + presigned read	Cloudflare R2
10	Transactional email (verify / reset)	frontend/src/lib/email/*	Auth flows	HKUST institutional SMTP relay (free)
11	Frontend hosting	frontend/	Always-on	Vercel
12	Backend + worker + Postgres	backend/ , railway.toml	Always-on (API pod + worker pod + PG plugin)	Railway

#	Surface	Files	Trigger	Pricing model
13	Canvas API	services/canvas_client.py , services/canvas_sync.py	Daily roster + file diff	Free (institutional Canvas)
14	Better Auth	frontend/src/app/api/auth/* + auth PG schema	All auth flows	Free (self- hosted)

2. Unit Prices (Q2 2026 reference)

All prices converted to HKD using **1 USD = 7.78 HKD**.

AI / model APIs

Model	Use	Input (HKD / M tokens)	Output (HKD / M tokens)	Source
DeepSeek V4 Flash	Primary LLM	0.871 (USD 0.112)	1.74 (USD 0.224)	OpenRouter
Gemini 2.5 Flash Lite	Fallback LLM	0.78 (USD 0.10)	3.11 (USD 0.40)	OpenRouter
Gemini 2.5 Flash	VLM (figures, OCR)	2.33 (USD 0.30)	19.45 (USD 2.50)	OpenRouter
openai/text-embedding-3-large	Embeddings	1.01 (USD 0.13)	—	OpenRouter
iFlytek (讯飞) Speech Evaluator	Chinese pronunciation grading (all four question types incl. long passage)	0.023 HKD/call (USD 0.003, ¥0.021 RMB) pay-as-you-go. Resource packs: USD 150/100K calls/yr (Pack A), USD 280/200K/yr (Pack B), USD 1,300/1M+/yr (Pack C)	—	iFlytek Open Platform pricing docs

Infrastructure

Vendor	Tier	Effective monthly (HKD)	Notes
Vercel	Pro (1 seat)	156 (USD 20)	Custom domain, analytics, function timeout >10s
Railway	Pro plan: \$20/mo credit floor. Plan fee is not additive when usage > floor.	(see §6.3)	API pod + worker pod + PG plugin all metered against the same \$20 floor
Railway compute rates	vCPU \$0.000463/min (~\$0.028/hr), RAM ~\$0.014/GB-hr, storage \$0.25/GB-mo, egress \$0.10/GB		
Cloudflare R2	Pay-as-you-go	12–50	Egress is free on R2; storage \$0.015/GB-mo
HKUST institutional SMTP	Institutional account (free)	0	Used for verification + reset emails; relay through HKUST IT services
OpenRouter	Pay-as-you-go, USD 100 prepaid	(modeled below)	Single credit pool for all LLM + embedding calls
iFlytek (讯飞)	Pay-as-you-go (no free-tier assumption in baseline)	(modeled below)	Chinese pronunciation grading. USD 0.003/call ≈ HKD 0.023. If HKUST qualifies for the enterprise free quota (200K/3-mo), the whole pronunciation line becomes near-zero — treat as upside. Resource packs (USD 150–1,300/yr) only beat PAYG above ~50K calls/year, which is far past our 4-month showcase volume

3. Per-Action Unit Cost (modeled, HKD) — using DeepSeek V4 Flash

Derived from prompt sizes in `services/generator.py` and chunk sizes in `services/chunker.py`. The +25% safety margin at the end absorbs fallback-model retries.

Action	Input tok	Output tok	Unit cost (HKD)
Quiz generation (1 call, 5–10 questions)	5,000	1,500	0.0070
Flashcard generation (1 set of 10 cards)	5,000	1,800	0.0075
Summary generation (course-wide)	10,000	2,500	0.0131
Pronunciation set generation (instructor, one-time)	4,000	2,000	0.0070
Concept extraction per sampled chunk	1,500	200	0.0017
Concept tagging per chunk	2,500	200	0.0025
RAG embedding per query	500	—	0.0005
Chunk embedding per chunk on upload	500	—	0.0005
Figure caption per image (VLM)	600 + image	400	0.009
Page-rescue OCR per scanned page (VLM)	500 + image	2,000	0.040
iFlytek pronunciation attempt (pay-as-you-go, USD 0.003/call)	—	—	0.023

One-time document-upload cost (instructor)

Typical course lecture deck = **80 chunks + 10 figures**. With concept extraction sampling 50 chunks:

```

chunks embedded:      80 × 0.0005 = 0.040
figure captions:      10 × 0.009  = 0.090
concept extraction:    50 × 0.0017 = 0.085
concept tagging:      80 × 0.0025 = 0.200
syllabus parse (per course, amortized over docs) = 0.020
_____
≈ 0.43 HKD / document

```

4. Usage Assumptions

User behaviour profiles for the live phase (Sep–Dec). Counts are **per user per month**. Pronunciation is treated as a **secondary feature**: many users try it once or twice and move on; quizzes + flashcards are the primary loop.

Profile	Share	Hrs/week	RAG queries	Quizzes gen	Flashcard sets	Summaries read	Pronunciation attempts
Light	50%	0.5	5	2	3	1	1
Moderate	35%	2.0	20	8	15	4	4
Heavy	15%	5.0	60	20	40	10	12

Weighted student average per month (50/35/15 mix): - RAG queries: 16.5 - Quizzes generated: 6.7 - Flashcard sets: 11.8 - Summaries read: 3.4 (cached, **free** after first gen) - Pronunciation attempts: 3.7

Per-student variable cost / month — V4 Flash + iFlytek pay-as-you-go (USD 0.003/call):

RAG embeds:	$16.5 \times 0.0005 = 0.008$
Quizzes:	$6.7 \times 0.0070 = 0.047$
Flashcards:	$11.8 \times 0.0075 = 0.089$
Pronunciation:	$3.7 \times 0.023 = 0.085$
	$\approx 0.23 \text{ HKD / student / month}$

iFlytek pay-as-you-go projection (4-month deployment, HKD):

Users	Attempts/mo	Cost/mo (HKD)	4-month cost (HKD)	4-month cost (USD)
10	33	0.8	3	0.4
50	181	4.2	17	2.2
100	363	8.4	33	4.3
200	722	16.6	67	8.6
500	1,806	41.5	166	21.4

PAYG beats resource packs at every scenario here — Pack A (USD 150/yr) only breaks even at ~50K calls/year, while our 500-user scenario uses ~7K calls in 4 months. If HKUST does qualify for the enterprise 200K-call/3-month free quota, all four-month costs drop to ~HKD 0.

Per-instructor variable cost / month (6 docs uploaded, 4 summaries, 4 pronunciation sets, 8 analytics views):

Doc uploads:	$6 \times 0.43 = 2.580$
Summaries:	$4 \times 0.0131 = 0.052$
Pronunciation sets:	$4 \times 0.0070 = 0.028$
Analytics narratives:	$8 \times 0.0131 = 0.105$
	$\approx 2.77 \text{ HKD / instructor / month}$

5. Phase A — Development & Staging (Jun – Aug, 3 months)

Internal team (~5 devs + 1 instructor seeding test content). No public traffic.

Line item	Monthly (HKD)	3-month (HKD)
Vercel Pro	156	468
Railway (API + worker + PG, dev shape, ~\$48/mo)	370	1,110
R2 storage (under 10 GB)	12	36
HKUST SMTP relay	0	0
AI surface — dev testing (~3K LLM calls + ~50 docs ingested, V4 Flash)	75	225
Phase A subtotal	613	1,839
+25% buffer		2,299

Phase A total: $\approx 2,300 \text{ HKD}$ (~USD 296 / RMB 2,100).

6. Phase B — Live Deployment (Sep – Dec, 4 months)

User mix: 1 instructor : 40 students:

Headline users	Students	Instructors
10	9	1
50	49	1
100	98	2
200	195	5
500	488	12

6.1 Variable AI cost per month (HKD) — V4 Flash + iFlytek PAYG (USD 0.003/call)

monthly_ai = students × 0.23 + instructors × 2.77

Users	Students × 0.23	Instructors × 2.77	Variable AI / month
10	2.07	2.77	5
50	11.27	2.77	14
100	22.54	5.54	28
200	44.85	13.85	59
500	112.24	33.24	145

6.2 Showcase-specific feature cost per month (HKD) — ADDITIVE on top of \$6.1

Forward-looking estimates with V4 Flash for the five new features.

Feature	Cost shape	Cost / student / mo	Cost / 100 users / mo
Placement test (5-group division)	1 LLM classification per student per semester	0.003	0.3
Mid-semester survey analysis	1 LLM analysis per student per semester	0.003	0.3
Instructor analytics dashboard	Mostly SQL (free). 1 LLM narrative per view, 8×/mo	(per instructor) 0.105	0.4
Department report	~10 reports/sem × ~0.026 HKD per longer report	(fixed)	1.6
Weak-point personalized quiz	Same shape as quiz gen; +4 extra/mo for moderate+heavy users	0.014	1.4
Subtotal			≈ 4 / 100 users / mo

If you want a **more elaborate adaptive placement test** (20–30 LLM-graded questions per student), bump the placement-test row from 0.003 to ~0.025 HKD/student/mo. At 500 users that adds ~12 HKD/month — negligible.

Scaling linearly:

Users	Showcase-features cost / mo (HKD)
10	0.4
50	2.0
100	4.0
200	8.0
500	20.0

6.3 Infrastructure cost per month (HKD)

Railway's Pro plan is a **\$20/mo credit floor**, not additive. Total Railway billing = `max(20, actual_usage)` . API pod (0.5 vCPU avg, 1 GB RAM) ≈ \$11/mo. Worker pod (0.3 vCPU avg, 1.5 GB RAM) ≈ \$17/mo. Postgres plugin (0.5 vCPU avg, 1 GB RAM, 4–10 GB storage) ≈ \$12–25/mo depending on scale.

Users	Vercel	Railway (API + Worker + PG combined)	R2	SMTP	Infra/mo
10	156	311 (\$40)	12	0	479
50	156	327 (\$42)	18	0	501
100	156	373 (\$48)	23	0	552
200	156	467 (\$60)	30	0	653
500	156	622 (\$80)	50	0	828

Levers to drop Railway further if needed: - **Move Postgres to Neon** (serverless, scale-to-zero) — saves ~\$10/mo at low traffic but adds latency. - **Sleep the worker pod during Jun–Aug staging** — Railway supports app sleep. Saves ~\$15/mo. - **Combine API + worker into one pod** behind asyncio (already how `app/main.py` runs the worker today in the lifespan). Loses isolation but saves a pod.

6.4 Phase B monthly + 4-month totals (HKD)

```
monthly_total = infra + variable_ai + showcase_features
```

Users	Infra	Var AI	Features	Monthly	× 4 mo	+25% buffer
10	479	5	0.4	484	1,937	2,421
50	501	14	2.0	517	2,068	2,585
100	552	28	4.0	584	2,336	2,920
200	653	59	8.0	720	2,880	3,600
500	828	145	20.0	993	3,972	4,965

7. 7-Month Headline Budget (HKD)

Phase A is constant. Phase B scales with deployment. **+25% buffer on the combined subtotal.**

Scenario (Sep–Dec users)	Phase A (Jun–Aug)	Phase B (Sep–Dec)	Subtotal	+25%	USD	RMB
10 users (closed pilot)	1,839	1,937	3,776	4,720	607	4,310
50 users (single course)	1,839	2,068	3,907	4,884	628	4,455
100 users (2–4 courses)	1,839	2,336	4,175	5,219	671	4,760
200 users (department pilot — recommended target)	1,839	2,880	4,719	5,899	758	5,380
500 users (upper bound / multi-dept showcase ceiling)	1,839	3,972	5,811	7,264	934	6,625

Recommended planning target: 200-user scenario at HKD 5,899 (≈ USD 758 / RMB 5,380). This is the highest-likely actual demand based on user guidance.

Ceiling provision (if conference traffic over-delivers): 500-user scenario at HKD 7,264 (≈ USD 934 / RMB 6,625).

Upside if iFlytek enterprise free tier is granted: drop pronunciation line entirely → 200-user scenario falls to HKD 5,809; 500-user scenario falls to HKD 7,049.

8. Dominant Cost Drivers & Levers

Ranking at the 200-user planning target (195 students + 5 instructors):

- Flashcard generation (~17 HKD/mo at 200 users, ~29% of variable AI).** 11.8 sets/mo × 0.0075 HKD per student. Largest line now that iFlytek's true price is roughly half what rev 6 assumed. Lever: cache previously generated sets keyed on chunk-set hash; partially supported by `flashcards.attempt` rows.
- iFlytek pronunciation grading (~17 HKD/mo at 200 users, ~28%).** 3.7 attempts/mo × 0.023 HKD per student (USD 0.003/call). Lever: confirm enterprise free-tier eligibility — if granted, this drops to zero. Also cap per-student daily attempts in `api/speech.py`.
- Instructor document processing (~13 HKD/mo at 200 users, ~22%).** Each instructor doc is ~0.43 HKD. Lever: cap `concept_extraction.py` sample size; respect `enable_figure_captions=false` in dev.
- Quiz generation (~9 HKD/mo at 200 users, ~15%).** 6.7 calls × 0.0070 HKD per student. Same caching lever applies.
- Postgres storage growth from pgvector.** A `vector(1536)` chunk is ~6 KB; a `vector(3072)` concept is ~12 KB. At 200 users × 20 courses × 100 chunks = ~2.4 GB of vectors per semester. Comfortable within Railway PG plugin defaults.
- VLM page-rescue OCR.** Already hard-capped at 40 pages per doc — keep that cap.

9. Recommended Paid Plans (final picks)

Vendor	Plan	Why
Vercel	Pro (1 seat, USD 20/mo)	Custom domain, function timeout >10s for proxy.ts auth handlers, deployment protection
Railway	Pro (USD 20/mo floor)	\$20 is a credit floor — actual bill = max(20, usage). Postgres plugin uses pgvector out-of-box
Cloudflare R2	Pay-as-you-go (no fixed fee)	Free egress is decisive for serving documents to students
HKUST institutional SMTP	Free institutional account	Verification + reset emails; ask HKUST IT for SMTP relay credentials
OpenRouter	Pay-as-you-go, USD 100 prepaid	Single credit pool for DeepSeek V4 Flash + Gemini Flash family + embeddings; one API key for all LLM surfaces
iFlytek (讯飞) Open Platform	Pay-as-you-go at USD 0.003/call. Top up ¥500 (≈ HKD 548 / USD 70) reserve before Sep go-live — this is ~3× the 500-user 4-month projected spend (HKD 166), so refill is only needed if usage massively over-runs. Also apply for enterprise free-quota (200K calls / 3 months) via HKUST institutional contact — if granted, the prepaid reserve sits unused as a pure safety float. Resource packs (USD 150–1,300/yr) are <i>not</i> economical at our scale	Chinese pronunciation grading incl. long passages. iFlytek bills daily against account balance; insufficient balance = immediate service outage , so the reserve is mandatory operational hygiene. Apply for enterprise approval by mid-August

Config change in backend/app/config.py : set openrouter_primary_model="deepseek/deepseek-v4-flash" (was deepseek/deepseek-chat-v3.2 per code audit). Keep Gemini 2.5 Flash Lite as the JSON-parse-failure fallback.

10. Files Critical to This Budget

If any of these change materially before Sep, re-run this budget:

- backend/app/config.py — model IDs (set primary to V4 Flash), enable_figure_captions , page_rescue_max_pages , rate limits
- backend/app/services/generator.py — every LLM-call entry point
- backend/app/services/embedder.py — batch size (currently 100)
- backend/app/services/concept_extraction.py — sampling cap (currently 200)
- backend/app/services/concept_tagger.py — per-chunk LLM tag
- backend/app/services/vlm.py — figure caption + page-rescue OCR
- backend/app/services/parser.py — Docling path (audio branch can be deleted alongside this budget)
- backend/app/services/speech.py — iFlytek (讯飞) integration. Remove any alternative-provider branches that may still exist
- frontend/src/lib/email/* — switch email transport from any prior provider to HKUST SMTP relay

- `backend/app/middleware/` — `/api/rag/*` rate limit (50 instr / 10 stu per hour) protects against runaway cost
- `railway.toml` — deployment shape

11. Verification Plan

Before submitting the budget to HKUST/PSC finance:

1. **Confirm OpenRouter list prices** for `deepseek/deepseek-v4-flash`, `google/gemini-2.5-flash`, `google/gemini-2.5-flash-lite`, and `openai/text-embedding-3-large` on the date of submission.
2. **Top up the iFlytek prepaid balance to ¥500 (≈ HKD 548) before Sep 1.** iFlytek's pay-as-you-go billing charges daily against account balance — if the balance hits zero, the API returns errors and pronunciation grading goes dark mid-showcase. Set a refill trigger (e.g. weekly Slack/email alert when balance falls below ¥200) and refill proactively if heavy usage materializes.
3. **Confirm iFlytek enterprise free-tier eligibility and quota** by contacting `openplatform@iflytek.com` with HKUST institutional credentials. Verify whether the 200K-call/3-month quota covers all four Speech-Evaluator question types (single character / word / sentence / **long passage**) under one meter, or whether long-passage grading counts as multiple billable calls. If the free tier is granted, the prepaid reserve becomes a pure safety float.
4. **Confirm HKUST SMTP relay availability** with HKUST IT; obtain credentials and verify deliverability through Better Auth's email transport.
5. **Run Phase A monthly numbers against actuals for July** once the first month of staging is billed — earliest reality-check before scaling assumptions to live.
6. **Reconcile per-student per-month variable cost** against the `api_usage` table at the end of October — ~6 weeks of live data is enough to refit the light/moderate/heavy mix.
7. **Re-run this plan in November** with actual Sep–Oct billing to forecast Dec spend with tighter error bars.

12. One-Page Summary for Finance

Meli (HKUST × PSC anniversary) — 7-month operating budget, Jun–Dec 2026

Three-month development phase (Jun–Aug) plus four-month live deployment (Sep–Dec). Mostly-student user mix (1 instructor : 40 students), capped at 500 users with 200 as the realistic target. LLM inference via **DeepSeek V4 Flash** (primary) + Gemini Flash family (fallback/VLM) on OpenRouter; Chinese pronunciation + long-passage grading via **iFlytek (讯飞)** Speech Evaluator, served entirely within the 200K-call/3-month enterprise free tier. Infrastructure: Vercel Pro, Railway Pro (API pod + worker pod + Postgres plugin), Cloudflare R2. Transactional email via HKUST institutional SMTP (free). Lecture-audio transcription out of scope.

Recommended provision (200-user target — highest likely): - HKD 5,899 (≈ USD 758 / RMB 5,380) including 25% safety buffer.

Ceiling provision (500-user upper bound): - HKD 7,264 (≈ USD 934 / RMB 6,625) including 25% safety buffer.

Closed-pilot floor (10 users): HKD 4,720.

Operational note: Provision a one-time **¥500 (≈ HKD 548)** prepaid balance with iFlytek before Sep 1 to avoid mid-showcase service outage. iFlytek bills daily against account balance and PAYG is USD 0.003/call (verified against official pricing docs); ¥500 covers ~3× the 500-user 4-month projected spend. If HKUST is granted iFlytek's enterprise 200K-call/3-month free quota, the pronunciation line drops to ~zero and the 200-user target falls to HKD 5,809.